

JS Crossdating Event Diagnosis Workflow

From format-preserving RWL data to one traceable event edit

STEP 1 DATA STATE AND REFERENCE CHRONOLOGY

Transparent RWL state

- Tucson short/long formats
- Raw baseline + working series
- Format-preserving read and write
- Target + same-site reference cores
- Year, zero and edit provenance

Reference chronology & side evidence

- Manual calendar-year chronology
- COFECHA-pass anchor master
- Spline detrending + AR prewhitening
- Leave-target-out reference
- PART 6 segment evidence
- Independent-core support

STEP 2 CONSTRAINED EVENT EVIDENCE

Lag topology

- Global sliding match
- COFECHA-like segment correlation
- Constrained integer-lag Viterbi path
- Direction, magnitude and plateaus
- Review windows: 5/7/9/13 years

Executable counterfactuals

- Virtual insert and delete
- Partial shifts: -2 to -100 years
- Whole-series shift grid
- Before/after correlations and residuals
- Executable operations only

Multi-core evidence ledger

- Master + independent-core votes
- Vote count, margin and concentration
- Terminal/fixed-side baseline
- Presence, operation and location types
- Provenance $\neq$ evidence weight

Missing ring

$g - 1 \rightarrow g$

False ring

$g + 1 \rightarrow g$

Partial shift

$g - q \rightarrow g; q \geq 2$

Whole-series shift

constant global g

STEP 3 JOINT ADJUDICATION: OPERATION $\times$ SHIFT $\times$ LOCATION

Linked hypothesis

Operation, lag path and evidence remain connected.

Joint adjudication

Fuse topology, counterfactuals and core votes.

Single product output

One executable edit + one focused review window.

Precise resolution

Resolve operation, shift and event location.

STEP 4 REVIEW, CONTROLLED COMMIT AND REDIAGNOSIS

Graph review

Inspect window and evidence

Edit preview

Select year or firstFixedYear

Confirm once

Apply one focused event

Refresh diagnosis

Rebuild on current working state

RwlEditor commit

Undo stack + operation log

FORMAT-PRESERVING

COUNTERFACTUAL

MULTI-CORE

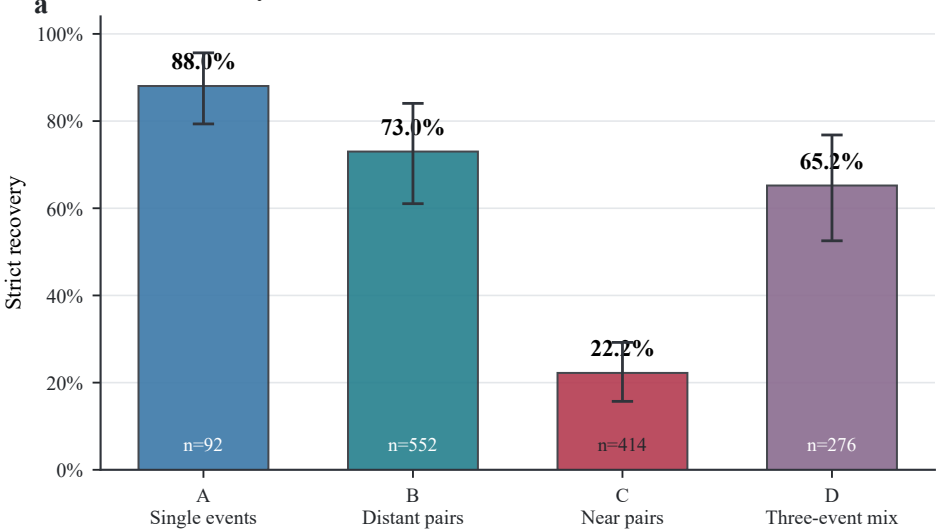
FULLY AUDITABLE

updated working state

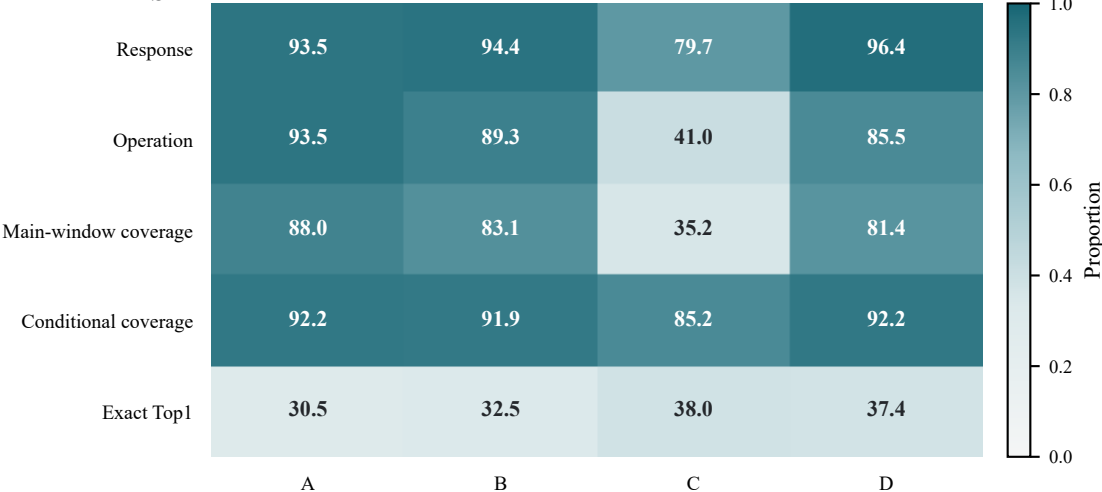
Cross-file performance of the dating recommendation module

23 unseen RWL files • 1,334 truth events • file-cluster bootstrap: 10,000 replicates

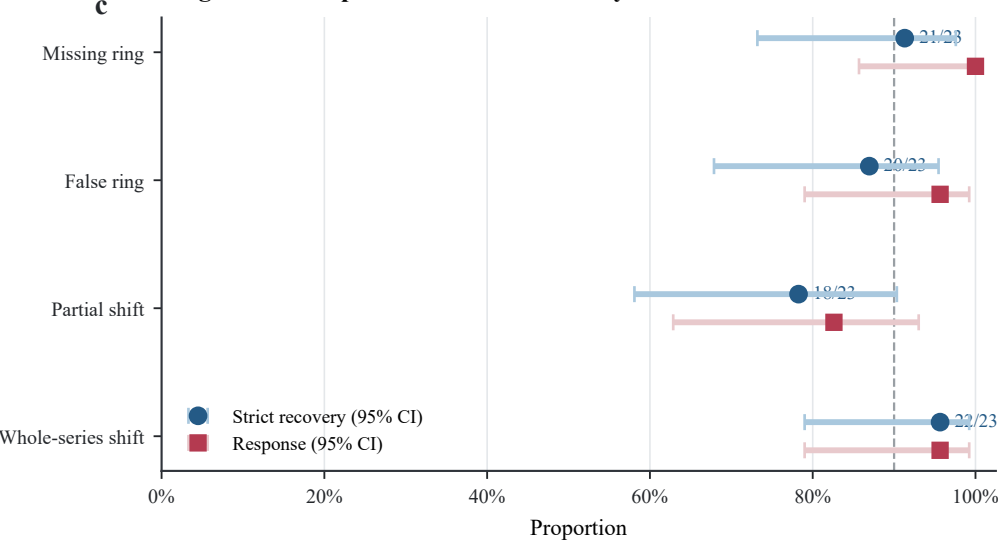
a Strict recovery across benchmark families



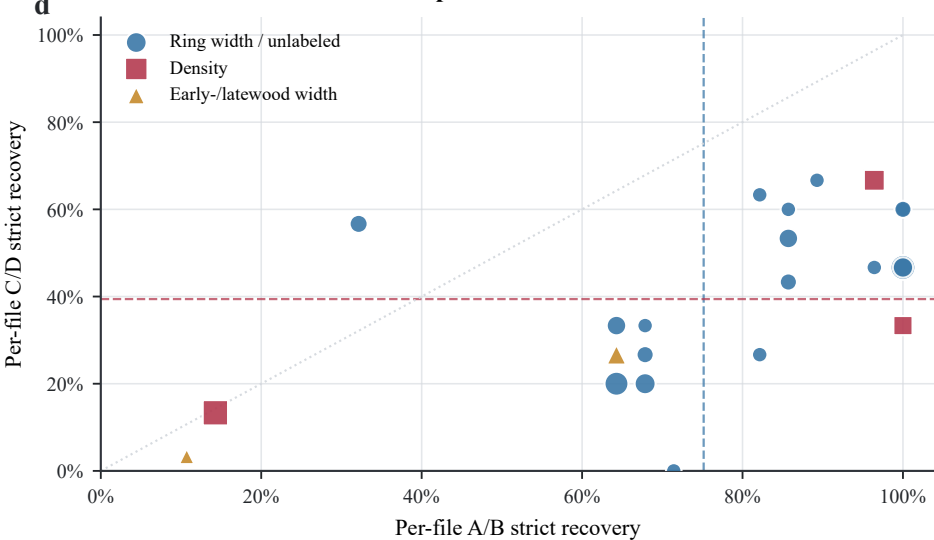
b Complementary performance dimensions (%)



c Single-event response and strict recovery



d Generalization across 23 independent RWL files



e



Crossdating events, ecological meaning and recognition evidence

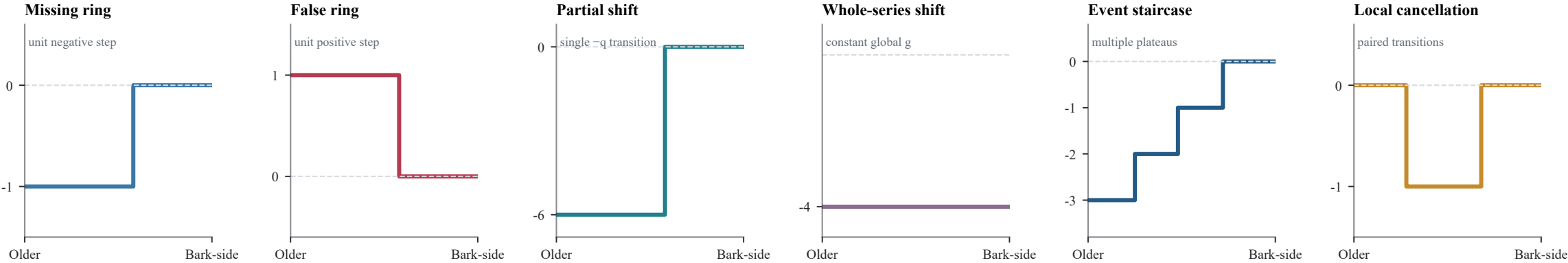
Lag signatures are read from older years toward the bark-side end.

Event and executable edit	Lag signature	Ecological / specimen meaning	Recognition evidence
Clean match Keep the series unchanged	Stable 0 (or a known global baseline)	Calendar alignment is retained across the measured series.	No persistent lag transition; edit counterfactuals provide no stable gain; reference cores agree on the baseline.
Missing ring Insert one calendar year	$g - 1 \rightarrow g$	A locally absent or extremely narrow ring, often associated with suppressed cambial growth under drought, cold, fire or insect stress.	A unit negative step, a superior insert-year counterfactual and concordant gains against the master and independent cores.
False ring Delete one calendar year	$g + 1 \rightarrow g$	An intra-annual density fluctuation or false boundary, including a ring divided twice during measurement.	A unit positive step, a superior delete-year counterfactual, residual support and consistent transition direction across references.
Partial shift Move the older block by $-q$ years	$g - q \rightarrow g$ $q \geq 2$	Localized decay, fracture, segment loss or a within-core measurement splice that displaces only part of the series.	One large local transition, a dominant range-shift counterfactual and agreement on the fixed-side breakpoint.
Whole-series shift Move the complete series by g years	Constant global g	A negative shift is consistent with tree death, missing bark or sapwood, or a bark-side break that removed outer years.	Global sliding match, a constant lag path, terminal-side evidence and a whole-series counterfactual converge on the same g .
Event staircase Apply sequential unit edits	$-n \dots -1 \rightarrow 0$ (or the positive mirror)	Several missing or false rings distributed through the series rather than one continuous displaced block.	Multiple intermediate lag plateaus, repeated unit counterfactual gains and stepwise recovery after each confirmed edit.
Adjacent paired events Resolve two local unit events	Short pulse; distal baseline restored	Nearby missing and false rings can cancel in long windows while retaining a distinct local disturbance signature.	Opposite local transitions, paired before/after evidence and matching votes on both sides of the short event interval.

Precise discrimination of complex crossdating patterns

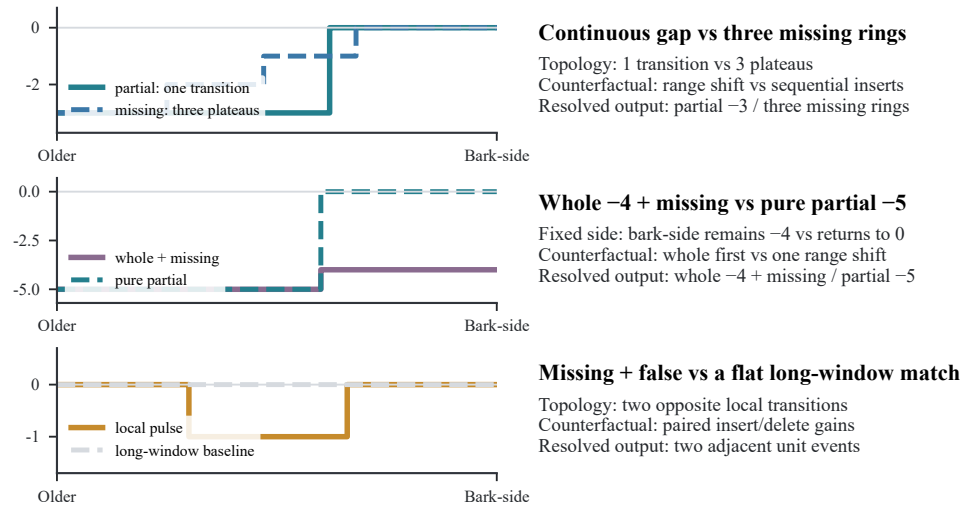
Lag topology + executable counterfactuals + independent-core evidence

a



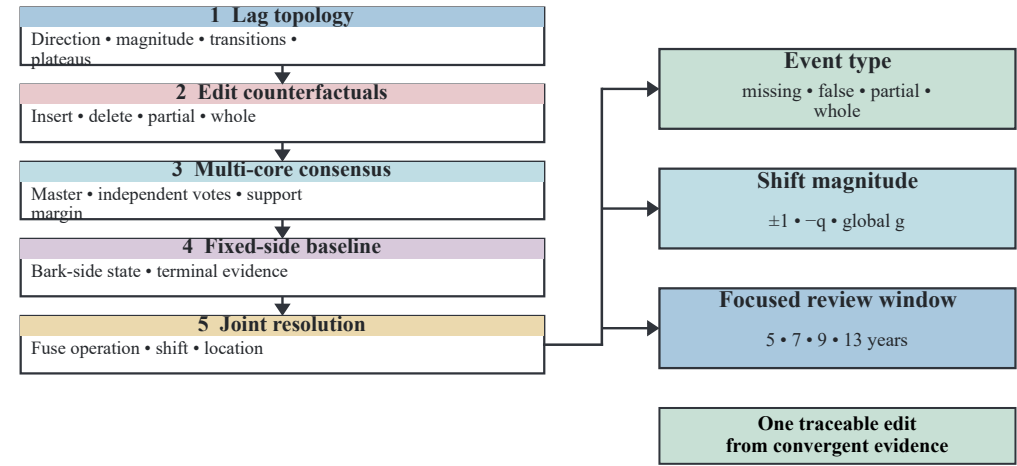
b

Three confusable patterns resolved by path structure



c

Evidence fusion produces a precise, executable result



d

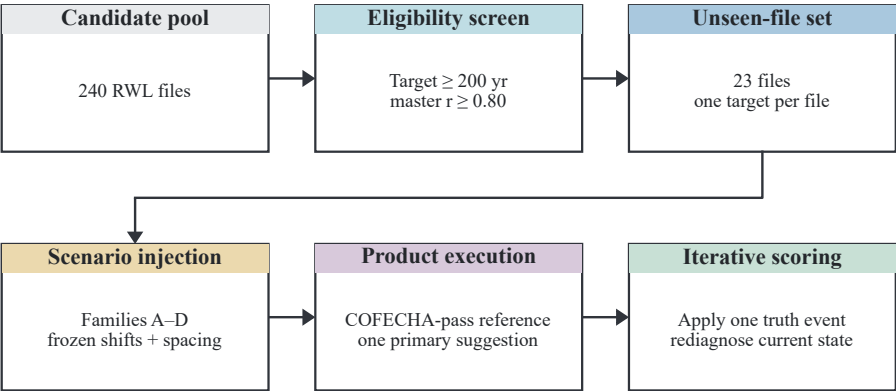
One-event iteration keeps every correction current and traceable



Cross-file benchmark design and capability coverage

23 unseen RWL files • four scenario families • file-cluster uncertainty • product-identical execution

a File-disjoint validation pipeline



b Four families span increasing event complexity

Family	Structure	Scen.	Cases	Truths	Capability
A	Clean + single events	5	115	92	Core operation
B	Distant pairs / whole + local	12	276	552	Separated recovery
C	Near local pairs (7 yr)	9	207	414	Local discrimination
D	Whole + two local events	4	92	276	Global + local

c Scenario coverage across diagnostic capabilities

A Single	YES	YES	YES	YES		
B Distant		YES	YES	YES		YES
C Near		YES	YES		YES	YES
D Mixed		YES	YES	YES	YES	YES
	Clean control	Unit edit	Range shift	Whole shift	Near events	Mixed events

d Reproducibility controls and update path

